[illegible]

"[illegible]" [illegible]

[illegible]

6 [illegible]

:$\mathcal{P}(\mathbb{N})$ [illegible] .F \col \mathbb{N} \to \mathbb{N} [illegible] ('[illegible] 30) .11

$$\langle A, B\rangle \in E \iff F[A] = F[B]$$

.$\mathcal{P}(\mathbb{N})/E$-[illegible] 2 [illegible] ('[illegible] 10) ([illegible])

.$E = Id_{\mathcal{P}(\mathbb{N})}$ [illegible] F-[illegible] ,[illegible] ('[illegible] 10) ([illegible])

.$\mathcal{P}(\mathbb{N})/E \sim \mathcal{P}(\mathbb{N})$ [illegible] F [illegible] ('[illegible] 10) ([illegible])

[illegible] – '[illegible] :[illegible]

[illegible] 2 [illegible]:

- [illegible]: $\emptyset$ ו-$\mathbb{N}$.
- [illegible]:

$$F[\emptyset] = \{F(x) \mid x \in \emptyset\} = \emptyset$$

- [illegible] $\mathbb{N}$ [illegible] ([illegible]):

$$F[\mathbb{N}] = \{F(x) \mid x \in \mathbb{N}\} \neq \emptyset$$

([illegible] $\mathbb{N}$ [illegible], [illegible]).

- [illegible] $y \in F[\mathbb{N}]$ ([illegible] $F(0)$) [illegible] $\emptyset$-[illegible], [illegible].
- [illegible] $F[\emptyset] \neq F[\mathbb{N}]$, [illegible] $\langle \emptyset, \mathbb{N} \rangle \notin E$.
- [illegible] $[\mathbb{N}]_E \neq [\emptyset]_E$, [illegible].

[illegible]: [illegible] 'ה – [illegible]

[illegible]: F [illegible] $\iff$ $E = Id_{\mathcal{P}(\mathbb{N})}$

$\Longleftarrow$ [illegible]: [illegible] F [illegible]. [illegible] $E = Id$.

- [illegible] $A, B \in \mathcal{P}(\mathbb{N})$ [illegible]-ש $\langle A, B\rangle \in E$,
- [illegible] $F[A] = F[B]$.
- [illegible] $A = B$:
  - [illegible] $x \in A$. [illegible] $F(x) \in F[A]$.
  - [illegible]-ש $F[A] = F[B]$, [illegible] $F(x) \in F[B]$.
  - [illegible], [illegible] $y \in B$ [illegible]-ש $F(x) = F(y)$.
  - [illegible]-ש F [illegible], [illegible]-ש $x = y$. [illegible] $x \in B$.
  - [illegible] $A \subseteq B$. [illegible] $B \subseteq A$. [illegible] $A = B$.

$\Longrightarrow$ [illegible]: [illegible] $E = Id$. [illegible] F [illegible].

- [illegible]-ש F [illegible].
- [illegible] $x \neq y$ [illegible]-ש $F(x) = F(y)$.
- [illegible] $A = \{x\}$ ו-$B = \{y\}$.
- $F[A] = \{F(x)\} = \{F(y)\} = F[B]$. [illegible] $\langle A, B\rangle \in E$.
- [illegible] $A \neq B \implies \{x\} \neq \{y\} \implies x \neq y$.
- [illegible] E-ש [illegible] Id-ל. [illegible].

[illegible]: [illegible] '[illegible] – [illegible] [illegible]

[illegible]: F [illegible]. [illegible]: $\mathcal{P}(\mathbb{N})/E \sim \mathcal{P}(\mathbb{N})$.

- [illegible] $\Phi : \mathcal{P}(\mathbb{N})/E \to \mathcal{P}(\mathbb{N})$ [illegible] $\Phi([A]_E) = F[A]$.
- [illegible]: [illegible] $[A] = [B]$ [illegible] $\langle A, B\rangle \in E$, [illegible] $F[A] = F[B]$. [illegible].
- [illegible]: [illegible] $\Phi([A]) = \Phi([B])$. [illegible] $F[A] = F[B]$. [illegible], $\langle A, B\rangle \in E$, [illegible] $[A] = [B]$.
- [illegible]: [illegible] $Y \subseteq \mathbb{N}$ ([illegible]).
  - [illegible] $A = F^{-1}[Y] = \{n \in \mathbb{N} \mid F(n) \in Y\}$.
  - [illegible] $\Phi([A]) = F[A] = F[F^{-1}[Y]]$.
  - [illegible] $F[F^{-1}[Y]] \subseteq Y$.
  - [illegible] F-[illegible], [illegible] $y \in Y$ [illegible] x [illegible] $F(x) = y$-[illegible], [illegible] $y \in F[F^{-1}[Y]]$.
  - [illegible] $F[A] = Y$, [illegible] $\Phi([A]) = Y$.
- [illegible]: Φ [illegible], [illegible].

12 [illegible]

:12 [illegible]

$.A^{B} \times A^{C} \sim A^{B \cup C}$ [illegible] $B \cap C = \emptyset$ [illegible] A, B, C [illegible] [illegible] [illegible] ('[illegible] 10) ([illegible])

$.T \sim \mathbb{N}$,$T \in D$ [illegible] $D \sim \mathbb{N}$-[illegible] [illegible] $\mathbb{N}$ [illegible] D [illegible] [illegible] [illegible] ('[illegible] 10) ([illegible])

[illegible]: [illegible] '[illegible] – [illegible] [illegible] [illegible]

[illegible]: $A^B \times A^C \sim A^{B \cup C}$ [illegible] $B \cap C = \emptyset$.

- [illegible] [illegible] \Phi \col A^B \times A^C \to A^{B \cup C}.
- [illegible] $A^B \times A^C$-[illegible] [illegible] [illegible] $\langle f, g \rangle$, [illegible] f \col B \to A [illegible]-g \col C \to A.
- [illegible] [illegible] $h = \Phi(\langle f, g \rangle)$ [illegible] h \col B \cup C \to A [illegible] [illegible]:

$$h(x) = \begin{cases} f(x) & x \in B \\ g(x) & x \in C \end{cases}$$

- [illegible] [illegible]: [illegible] -$B \cap C = \emptyset$, [illegible] $x \in B \cup C$ [illegible] [illegible] [illegible] [illegible], [illegible] $h(x)$ [illegible] [illegible] [illegible].
- [illegible]: [illegible] $\Phi(\langle f_1, g_1 \rangle) = \Phi(\langle f_2, g_2 \rangle)$. [illegible] $h_1 = h_2$.
  - [illegible] $x \in B$, $f_1(x) = h_1(x) = h_2(x) = f_2(x)$, [illegible] $f_1 = f_2$.
  - [illegible] $x \in C$, $g_1(x) = h_1(x) = h_2(x) = g_2(x)$, [illegible] $g_1 = g_2$.
  - [illegible] $\langle f_1, g_1 \rangle = \langle f_2, g_2 \rangle$.
- [illegible]: [illegible] $h \in A^{B \cup C}$. [illegible] $f = h|_B$ -[illegible] $g = h|_C$. [illegible] $\Phi(\langle f, g \rangle) = h$.
- [illegible] [illegible] [illegible], [illegible] [illegible] [illegible] [illegible].

□□□□□□: □□□□ □' – □□□□□□ □□□□□□□□□□ □□ □□□□□□□□□

□"□: □□□□□□ □□□□□□ D □□ $\mathbb{N}$ □□ □-$\aleph_0 = |D|$ □□□□ $T \in D$, $|T| = \aleph_0$.

- □□□□□□ □□□□□□□□□ □□□□□□□ □□ □□□□□□ (□□ □□ □□□□□□□□ □□□□□□ □□□□□) J \col \mathbb{N} \times \mathbb{N} \to \mathbb{N}.
- □□□□□□ $\{\{k\} \times \mathbb{N} \mid k \in \mathbb{N}\}$ □□□ □□□□□□ □□ $\mathbb{N} \times \mathbb{N}$. (□□ □□□ □□□□□ □□□□□ □□□□□□ □□□□□□ □□□ k).
- □□□□□□ □□ □□□□□□ □□□□□□□ T_k □□□□□□□□ □□ □□□□□□□ □□□□□ □□□ J:

$$T_k = \{J(k,n) \mid n \in \mathbb{N}\}$$

- □□□□□□ □□ □□□□□□□ $D = \{T_k \mid k \in \mathbb{N}\}$.
- □□□□□□□:
  1. □□□□□□□ □-J □□□□□, □□□□□□□ □□□□□□□ □□□□□□□ □□□□□□□ □-$\mathbb{N}$.
  2. □□□□□□ □□□□□□□: □□□□□□□□□ $k \mapsto T_k$ □□□ □□"□, □□□ $|D| = |\mathbb{N}| = \aleph_0$.
  3. □□□□□□ □□□□□□□□: □□□ k, □□□□□□□□□ $n \mapsto J(k,n)$ □□□ □□"□ □□□ □-$\mathbb{N}$ □-T_k. □□□ $|T_k| = \aleph_0$.

13 [illegible]

13 [illegible]: [illegible] R [illegible] A [illegible] $T = \{S \subseteq A \times A \mid S \text{ [illegible]} \wedge R \subseteq S\}$.

(א) (10 [illegible]) [illegible] $\bigcap T \in T$.

(ב) (20 [illegible]) [illegible] A [illegible] $i \in \mathbb{N}$:

- $R_0 = R$
- $R_{i+1} = R_i \cup \{\langle x, y\rangle \mid \exists z(\langle x, z\rangle \in R_i \wedge \langle z, y\rangle \in R_i)\}$

[illegible] $\bigcap T = \bigcup\{R_i \mid i \in \mathbb{N}\}$.

[illegible]: [illegible] – [illegible]

[illegible]: $\bigcap T \in T$. [illegible], [illegible] T-[illegible] [illegible] R.

1. [illegible] R:
   - [illegible] $S \in T$, [illegible] $R \subseteq S$ ([illegible] T).
   - [illegible] $\bigcap T = \bigcap\{S \mid S \in T\} \subseteq R$.
2. [illegible]:
   - [illegible] $\langle x, y\rangle \in \bigcap T$ [illegible]-$\langle y, z\rangle \in \bigcap T$.
   - [illegible] $S \in T$, $\langle x, y\rangle \in S$ [illegible] $\langle y, z\rangle \in S$.
   - [illegible] $S \in T$ [illegible], [illegible] $\langle x, z\rangle \in S$ [illegible] $S \in T$.
   - [illegible] $\langle x, z\rangle \in \bigcap T$.
   - [illegible], [illegible] $\bigcap T$ [illegible].

[illegible]: $\bigcap T \in T$.

(1 [illegible]) [illegible] - '[illegible] [illegible] :[illegible]

[illegible] $R^* = \bigcup_{i\in\mathbb{N}} R_i$. [illegible]: $R^* = \bigcap T$.

[illegible] 1: $\bigcap T \subseteq R^*$

- [illegible] $R^* \in T$ ([illegible] R). [illegible], [illegible] $R^* \subseteq \bigcap T$ [illegible] ([illegible] $\bigcap T$ [illegible] T-[illegible]).
- [illegible]: $R = R_0 \subseteq \bigcup R_i = R^*$.
- [illegible]:
  - [illegible] $\langle x, y\rangle \in R^*$-[illegible] $\langle y, z\rangle \in R^*$.
  - [illegible] i, j [illegible] $\langle x, y\rangle \in R_i$-[illegible] $\langle y, z\rangle \in R_j$.
  - [illegible] $k = \max(i, j)$. [illegible] ($R_n \subseteq R_{n+1}$), [illegible] R_k-[illegible].
  - [illegible] R_{k+1}, [illegible] "[illegible]" R_k-[illegible], [illegible] $\langle x, z\rangle$ [illegible] R_{k+1}-[illegible].
  - [illegible] $\langle x, z\rangle \in R_{k+1} \subseteq R^*$.
- [illegible] $R^* \in T$, [illegible] $\bigcap T \subseteq R^*$.

□□□□□□: □□□□ '□ - □□□□□ □□□□□□□□□□□ (□□□ 2)

□□□□□ 2: $R^* \subseteq \bigcap T$

- □□□□□ □□□□□□ □□□□ $S \in T$, □□□□□□ $R^* \subseteq S$.
- □□□□□ □□□□□□ □□□□ i, $R_i \subseteq S$ (□□ □□ □□□□□□ □□□□ □-S).
- □□□□□ □□□□□□□□□□ □□ i:
  - □□□□ ($i = 0$): $R_0 = R$. □□□□ $S \in T \implies R \subseteq S$. □□□□□□.
  - □□□ ($i \to i + 1$): □□□□ $R_i \subseteq S$.
    - □□□ $\langle x, y\rangle \in R_{i+1}$.
    - □□□□□□ □': $\langle x, y\rangle \in R_i$. □□□□□ □□□□□□□□□□, □-$S$.
    - □□□□□□ □': □□□□ z □□ □-$\langle x, z\rangle \in R_i$ □-$\langle z, y\rangle \in R_i$.
    - □□□□□ □□□□□□□□□□, $\langle x, z\rangle \in S$ □-$\langle z, y\rangle \in S$.
    - □□□□□□ □-S □□□□□□□□□, $\langle x, y\rangle \in S$.
  - □□□ $R_{i+1} \subseteq S$.
- □□□□ $R^* \subseteq S$ □□□ $S \in T$, □□□□ $R^* \subseteq \bigcap T$.

□□□□□ 14

□□□□□ 14: (□□' 32) □□□□ A □-B □□□□□□□ □□□□ f \col A \to B. □□□□□ □□□□□□□□□ □□□□: g \col \mathcal{P}(B) \to \mathcal{P}(A) □□
□□□□ $Y \in \mathcal{P}(B)$,

$$g(Y) = f^{-1}[Y]$$

(□) (□□' 16) □□□□□□□ □-f □□ B □□"□ g □□"□.

(□) (□□' 16) □□□□□□□ □-f □□"□ □"□□ g □□ $\mathcal{P}(A)$.

[illegible]: [illegible] – [illegible] (1 [illegible])

[illegible]: f [illegible] $\iff$ g [illegible].

$\Leftarrow$ [illegible]: [illegible] f [illegible]. [illegible] g-[illegible].

- [illegible] $Y_1, Y_2 \subseteq B$ [illegible] $g(Y_1) = g(Y_2)$-[illegible].
- [illegible] $f^{-1}[Y_1] = f^{-1}[Y_2]$.
- [illegible] f [illegible]: $f[f^{-1}[Y_1]] = f[f^{-1}[Y_2]]$.
- [illegible] f-[illegible], [illegible] $Y \subseteq B$ [illegible] Y $f[f^{-1}[Y]] = Y$:
  - [illegible] $f[f^{-1}[Y]] \subseteq Y$.
  - [illegible], [illegible] $y \in Y$, [illegible] x [illegible] $f(x) = y$-[illegible] ([illegible] f [illegible]).
  - [illegible] $x \in f^{-1}[Y]$ [illegible] $y \in f[f^{-1}[Y]]$.
- [illegible] $Y_1 = Y_2$.
- [illegible] g [illegible].

$\Longrightarrow$ [illegible]: [illegible] g [illegible]. [illegible] f-[illegible].

- [illegible] f-[illegible]. [illegible] $y \in B \setminus \mathrm{Im}(f)$.
- [illegible] $Y_1 = \{y\}$ [illegible] $Y_2 = \emptyset$-[illegible].
- $g(Y_1) = f^{-1}[\{y\}]$. [illegible] y-[illegible], [illegible] x [illegible] $f(x) = y$, [illegible] $f^{-1}[\{y\}] = \emptyset$.
- $g(Y_2) = f^{-1}[\emptyset] = \emptyset$.
- [illegible] $g(Y_1) = g(Y_2) = \emptyset$, [illegible] $Y_1 \neq Y_2$.
- [illegible] g-[illegible]. [illegible] f [illegible].

(2 [illegible]) [illegible] [illegible] – ׳[illegible] [illegible] :[illegible]

[illegible]: f [illegible] $\iff$ g [illegible].

$\Leftarrow$ [illegible]: [illegible] f [illegible]. [illegible] g-[illegible] $\mathcal{P}(A)$.

- [illegible] $X \subseteq A$. [illegible] $Y \subseteq B$ [illegible] X-[illegible] $g(Y) = X$.
- [illegible] $Y = f[X] = \{f(x) \mid x \in X\}$
- [illegible] $g(Y) = f^{-1}[f[X]]$.
- [illegible] f-[illegible], [illegible] $X \subseteq A$ [illegible] $f^{-1}[f[X]] = X$:
  - [illegible] $X \subseteq f^{-1}[f[X]]$.
  - [illegible], [illegible] $z \in f^{-1}[f[X]]$, [illegible] $f(z) \in f[X]$.
  - [illegible] $f(z) = f(x)$ [illegible] $x \in X$.
  - [illegible] f-[illegible], $z = x \in X$.
- [illegible] $g(Y) = X$.
- [illegible] X [illegible], [illegible] g [illegible].

$\Rightarrow$ [illegible]: [illegible] g [illegible]. [illegible] f-[illegible].

- [illegible] f-[illegible]. [illegible] $x_1 \neq x_2$ [illegible]-$f(x_1) = f(x_2) = y$.
- [illegible] $X = \{x_1\}$.
- [illegible] g-[illegible], [illegible] $Y \subseteq B$ [illegible] X-[illegible] $g(Y) = X$, [illegible] $f^{-1}[Y] = \{x_1\}$.
- [illegible] $y \in Y$: [illegible] $x_1 \in f^{-1}[Y]$ [illegible] $x_2 \in f^{-1}[Y]$ ([illegible] $f(x_2) = y$). [illegible] $f^{-1}[Y]$ [illegible] x_1. [illegible] ($x_1 \neq x_2$).
- [illegible] $y \notin Y$: [illegible] $x_1 \notin f^{-1}[Y]$ ([illegible] $f(x_1) = y$). [illegible]-$f^{-1}[Y] = \{x_1\}$.
- [illegible], [illegible] f [illegible].

[illegible] 40)

7. [illegible] (40 [illegible]): [illegible] $\langle A,R\rangle,\langle B,S\rangle$ [illegible]. [illegible] $\langle A,R\rangle \oplus \langle B,S\rangle = \langle C,T\rangle$ [illegible]:
$C=(\{0\}\times A)\cup(\{1\}\times B)$ [illegible] T [illegible]:

$$T=\{(\langle 0,x\rangle,\langle 0,y\rangle)\mid xRy\}\cup\{(\langle 1,x\rangle,\langle 1,y\rangle)\mid xSy\}\cup\{(\langle 0,x\rangle,\langle 1,y\rangle)\mid x\in A\wedge y\in B\}$$

([illegible]) (20 [illegible]) [illegible] $\langle A,R\rangle,\langle B,S\rangle$ [illegible], [illegible] $\langle C,T\rangle$ [illegible].

([illegible]) (20 [illegible]) [illegible]: [illegible] $(\mathbb{N},\leq^{\mathbb{N}})\cong(\{0\},\{(0,0)\})\oplus(\mathbb{N},\leq^{\mathbb{N}})$ [illegible] $(\{0\},\{(0,0)\})\oplus(\mathbb{N},\leq^{\mathbb{N}})\ncong(\mathbb{N},\leq^{\mathbb{N}})$

[illegible]: [illegible] [illegible] ([illegible] [illegible])

1. [illegible]:
   - [illegible]: [illegible] $z \in C$.
     - [illegible] $z = \langle 0, x \rangle$: [illegible] R-[illegible] [illegible], xRx, [illegible] zTz.
     - [illegible] $z = \langle 1, y \rangle$: [illegible] S-[illegible] [illegible], ySy, [illegible] zTz.
   - [illegible]-[illegible]: [illegible] uTv [illegible]-vTu.
     - [illegible] $u, v \in \{0\} \times A$: [illegible] $u = \langle 0, x \rangle, v = \langle 0, y \rangle$. [illegible] T, xRy [illegible]-yRx. [illegible]-R [illegible]-[illegible] $u = v \implies x = y \implies$.
     - [illegible] $u, v \in \{1\} \times B$: [illegible]"[illegible] S, [illegible] $u = v$.
     - [illegible] [illegible] ([illegible]-$\{0\} \times A$ [illegible]-$\{1\} \times B$): [illegible] [illegible] uTv [illegible] vTu, [illegible] T [illegible] [illegible] 0 [illegible] 1, [illegible] [illegible] 1 [illegible] 0.
   - [illegible]: [illegible] uTv [illegible]-vTw.
     - [illegible] $u, v, w \in \{0\} \times A$: [illegible] R.
     - [illegible] $u, v, w \in \{1\} \times B$: [illegible] S.
     - [illegible] $u \in \{0\} \times A$ [illegible]-$w \in \{1\} \times B$: [illegible] T ([illegible] [illegible]), [illegible] uTw [illegible] ([illegible] 0-[illegible] [illegible] 1-[illegible]), [illegible] [illegible]-v.
     - [illegible] [illegible] ([illegible] $u \in \{1\}, v \in \{0\}$ [illegible] uTv).
2. [illegible] ([illegible]): [illegible] $u, v \in C$ [illegible].
   - [illegible] $u, v \in \{0\} \times A$: [illegible] $u = \langle 0, x \rangle, v = \langle 0, y \rangle$. [illegible] R-[illegible], [illegible] xRy [illegible] yRx. [illegible] uTv [illegible] vTu.
   - [illegible] $u, v \in \{1\} \times B$: [illegible]"[illegible] S.
   - [illegible] u, v-[illegible] $\theta \times A$-[illegible] $1 \times B$: [illegible] $u \in \{0\} \times A$ [illegible]-$v \in \{1\} \times B$. [illegible] T ([illegible] [illegible]), [illegible] uTv.
   - [illegible], [illegible] T-[illegible] [illegible].

[illegible]: [illegible] [illegible] ([illegible] [illegible])

[illegible] '[illegible]: [illegible]

1. [illegible] -[illegible] $(\mathbb{N}, \leq^{\mathbb{N}}) \cong (\mathbb{N}, \leq^{\mathbb{N}}) \oplus (\{0\}, \{(0,0)\})$:

- [illegible] $C = (\{0\} \times \{0\}) \cup (\{1\} \times \mathbb{N})$.
- [illegible] [illegible] $f : C \to \mathbb{N}$ [illegible] [illegible]:

$$f(z) = \begin{cases} 0 & z = \langle 0, 0 \rangle \\ n+1 & z = \langle 1, n \rangle \end{cases}$$

- [illegible] [illegible]: [illegible] [illegible] f [illegible] [illegible] [illegible] [illegible] 0-[illegible]-0, [illegible] [illegible] [illegible] [illegible] 1-[illegible], [illegible] [illegible] [illegible].
- [illegible] [illegible]:
  - [illegible] [illegible] [illegible] [illegible] 1: $n+1 < m+1 \iff n < m$.
  - [illegible] [illegible] [illegible]: $\langle 0, 0 \rangle$ [illegible] [illegible] [illegible] -[illegible]$\langle 1, n \rangle$ ([illegible] [illegible] [illegible]). [illegible] $0 < n+1$ [illegible] $n \in \mathbb{N}$.
- [illegible] [illegible] [illegible].

2. [illegible] -[illegible] $(\mathbb{N}, \leq^{\mathbb{N}}) \ncong (\mathbb{N}, \leq^{\mathbb{N}}) \oplus (\{0\}, \{(0,0)\})$:

[illegible] $D = (\{0\} \times \mathbb{N}) \cup (\{1\} \times \{0\})$

[illegible]

[illegible]: [illegible] ($\mathcal{P}(\mathbb{N})$ [illegible] $\mathcal{P}(\mathbb{N}) \setminus \{\emptyset\}$), [illegible]:

- [illegible],[illegible],[illegible],[illegible]-[illegible]

[illegible]:

1. [illegible] $\mathcal{P}(\mathbb{N})$: $A \cap B = \emptyset \iff AR_1B$ ([illegible])
2. [illegible] $\mathcal{P}(\mathbb{N}) \setminus \{\emptyset\}$: $\min(A) = \min(B) \iff AR_2B$ ([illegible])
3. [illegible] $\mathcal{P}(\mathbb{N})$: $|A| \leq |B| \iff AR_3B$ ([illegible])
4. [illegible] $\mathcal{P}(\mathbb{N}) \setminus \{\emptyset\}$: $\min(A) > \min(B) \iff AR_4B$ ([illegible])
5. [illegible] $\mathcal{P}(\mathbb{N})$: $1 \in A \cap B \iff AR_5B$ ([illegible] 1)
6. [illegible] $\mathcal{P}(\mathbb{N})$: $A \subseteq B \iff AR_6B$ ([illegible])
7. [illegible] $\mathcal{P}(\mathbb{N})$: $A \cap B \neq \emptyset \iff AR_7B$ ([illegible])
8. [illegible] $\mathcal{P}(\mathbb{N})$: $A \cup B = \mathbb{N} \iff AR_8B$ ([illegible])
9. [illegible] $\mathcal{P}(\mathbb{N})$: $A \times B = \emptyset \iff AR_9B$ ([illegible])
10. [illegible] $\mathcal{P}(\mathbb{N})$: $|A \cap B| < \aleph_0 \iff AR_{10}B$ ([illegible])

[illegible] [illegible] [illegible] (1 [illegible])

1. $AR_1B \iff A \cap B = \emptyset$

- [illegible]: . ([illegible] $A = \{1\}$, $A \cap A \neq \emptyset$).
- [illegible]: . ($A \cap B = B \cap A$).
- [illegible]: . ($A = \{1\}, B = \{2\}, C = \{1\}$. AR_1B [illegible]-BR_1C [illegible] [illegible] AR_1C).
- [illegible]-[illegible]: . ($\{1\}R_1\{2\}$ [illegible]-$\{2\}R_1\{1\}$ [illegible] [illegible]).

2. $AR_2B \iff \min(A) = \min(B)$

- [illegible]: . [illegible] [illegible] [illegible] [illegible].
- [illegible]: .
- [illegible]: .
- [illegible]-[illegible]: . ($\{1,2\}R_2\{1,3\}$ [illegible] [illegible]).
- [illegible]: [illegible], [illegible] [illegible] [illegible].

[illegible] (2 [illegible])

3. $AR_3B \iff |A| \leq |B|$

- [illegible]: .
- [illegible]: . ([illegible]).
- [illegible]: . ($\leq$-[illegible]).
- [illegible]-[illegible]: . ($\{1\}$-[illegible] $\{2\}$ [illegible]).

4. $AR_4B \iff \min(A) > \min(B)$

- [illegible]: . ($x > x$ [illegible]).
- [illegible]: .
- [illegible]: .
- [illegible]-[illegible]: ([illegible]). [illegible], [illegible] "[illegible]" [illegible].

[illegible]: [illegible] [illegible] ([illegible] 3)

5. $AR_5B \iff 1 \in A \cap B$

- [illegible]: ([illegible] 1). [illegible] 1.
- [illegible]: .
- [illegible]: .
  - [illegible] ARB [illegible] $1 \in A$ [illegible]-$1 \in B$.
  - [illegible] BRC [illegible] $1 \in B$ [illegible]-$1 \in C$.
  - [illegible] $1 \in A$-[illegible] $1 \in C$ [illegible] $1 \in A \cap C$, [illegible] ARC.
- [illegible]-[illegible]: .

6. $AR_6B \iff A \subseteq B$

- [illegible]: .
- [illegible]: .
- [illegible]: .
- [illegible]-[illegible]: . ([illegible]).

[illegible]: [illegible] [illegible] (4 [illegible])

7. $AR_7B \iff A \cap B \neq \emptyset$

- [illegible]: $(\emptyset \cap \emptyset = \emptyset)$. [illegible]
- [illegible]: .
- [illegible]: $(A = \{1,2\}, B = \{2,3\}, C = \{3,4\}$ - [illegible], [illegible] $A \cap C = \emptyset)$.
- [illegible]-[illegible]: .

8. $AR_8B \iff A \cup B = \mathbb{N}$

- [illegible]: $(A \cup A = A \neq \mathbb{N}$ [illegible]). [illegible] $A = \mathbb{N}$.
- [illegible]: .
- [illegible]: $(A = \{1\}^c, B = \{1\}, C = \{1\}^c$. $A \cup B = \mathbb{N}, B \cup C = \mathbb{N}$, [illegible] $A \cup C = A \neq \mathbb{N})$.
- [illegible]-[illegible]: .

[illegible]: [illegible] [illegible] ([illegible] 5)

9. $AR_9B \iff A \times B = \emptyset$

- [illegible]: [illegible] $A = \emptyset$-[illegible] [illegible] $B = \emptyset$.
- [illegible]: . ($A \times A \neq \emptyset$ [illegible]). [illegible].
- [illegible]: . ($A = \emptyset \lor B = \emptyset \iff B = \emptyset \lor A = \emptyset$).
- [illegible]: ([illegible]). [illegible] $A = \{1\}, B = \emptyset, C = \{1\}$. [illegible] ARB -[illegible] BRC. [illegible] $A \times C \neq \emptyset$.
- [illegible]-[illegible]: .

10. $AR_{10}B \iff |A \cap B| < \aleph_0$

- [illegible]: . ([illegible] A [illegible], [illegible]). [illegible].
- [illegible]: .
- [illegible]: . $A = C = \mathbb{N}$, $B = \emptyset$.
- [illegible]-[illegible]: .

[illegible]

[illegible]: [illegible] $f : \mathbb{N} \to \mathcal{P}(\mathbb{N})$ [illegible]:

$$f(x) = \{n \in \mathbb{N} \mid 2^{n-1} \leq x\}$$

[illegible] T [illegible] $\mathbb{N}$ [illegible]:

$$xTy \iff f(x) = f(y)$$

[illegible]:

[illegible]. [illegible] T [illegible] $\mathbb{N}$.

[illegible]. [illegible] n [illegible]:

$$[2^n]_T = \{m \in \mathbb{N} \mid 2^n \leq m < 2^{n+1}\}$$

([illegible] $[2^n]_T$ [illegible] 2^n).

[illegible]: [illegible] [illegible] [illegible] [illegible]"[illegible] [illegible]

[illegible] T-[illegible] [illegible] [illegible]:

- [illegible] (Kernel equivalence).
- [illegible]: [illegible] x, $f(x) = f(x)$ [illegible] xTx.
- [illegible]: [illegible] xTy [illegible] $f(x) = f(y)$, [illegible] $f(y) = f(x)$ [illegible] yTx.
- [illegible]: [illegible] xTy [illegible]-yTz, [illegible] $f(x) = f(y)$ [illegible]-$f(y) = f(z)$, [illegible] $f(x) = f(z)$ [illegible] xTz.

[illegible] $[2^n]_T$:

$\{m \in \mathbb{N} \mid 2^n \leq m < 2^{n+1}\} \subseteq [2^n]$:

- [illegible] $m \in \{m \in \mathbb{N} \mid 2^n \leq m < 2^{n+1}\}$, [illegible] $2^n \leq m < 2^{n+1}$.
- [illegible] $k \leq n+1$: $2^{k-1} \leq 2^n \leq m$, [illegible] $\{0, ..., n+1\} \subseteq f(m)$.
- [illegible] $k \geq n+2$: $2^{k-1} \geq 2^{n+1} > m$, [illegible] $k \notin f(m)$.
- [illegible] $f(m) = \{0, ..., n+1\} = f(2^n)$, [illegible] $m \in [2^n]$.

$[2^n] \subseteq \{m \in \mathbb{N} \mid 2^n \leq m < 2^{n+1}\}$:

- [illegible] $m \in [2^n]_T$, [illegible] $f(m) = f(2^n) = \{0, ..., n+1\}$.
- [illegible] $n+1 \in f(m)$, [illegible] $2^n \leq m \iff 2^{(n+1)-1} \leq m$
- [illegible] $n+2 \notin f(m)$, [illegible] $2^{n+1} > m \iff 2^{(n+2)-1} > m$
- [illegible]"[illegible] $2^n \leq m < 2^{n+1}$, [illegible] $m \in \{m \in \mathbb{N} \mid 2^n \leq m < 2^{n+1}\}$

[illegible]

[illegible]: $(Y,\leq)$ [illegible]. $A = Y^X$ [illegible]. A [illegible]:

1. [illegible]: $\forall x \in X(f(x) \leq g(x)) \iff fR_1g$
2. [illegible]: $\exists x \in X(f(x) \leq g(x)) \iff fR_2g$

[illegible]: R_1, R_2 [illegible] A.

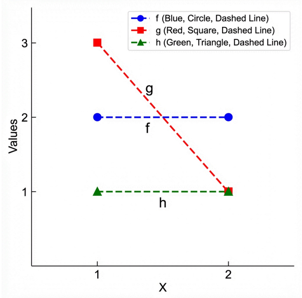
[illegible]: [illegible] [illegible] [illegible] [illegible]

1. [illegible] "[illegible]" (R_1): [illegible].

- [illegible]: [illegible] x, $f(x) \leq f(x)$ [illegible] fR_1f.
- [illegible]: [illegible] $\forall x(f(x) \leq g(x))$ [illegible] $\forall x(g(x) \leq f(x))$, [illegible] Y, [illegible] x [illegible] $f(x) = g(x)$, [illegible] $f = g$.
- [illegible]: [illegible] fR_1g [illegible] gR_1h.
  - [illegible] x: $f(x) \leq g(x)$ [illegible] $g(x) \leq h(x)$.
  - [illegible] Y [illegible] x: $f(x) \leq h(x)$.
  - [illegible] fR_1h.

2. [illegible] "[illegible]" (R_2): [illegible].

- [illegible]: [illegible] ([illegible] x [illegible], [illegible] $X \neq \emptyset$).
- [illegible]: [illegible].
  - [illegible]: $Y = \mathbb{N}$, $X = \{1, 2\}$, $f(1) = g(2) = 2$, $f(2) = g(1) = 1$.
  - $f(1) \leq g(1)$ [illegible] $g(2) \leq f(2)$ [illegible] fR_2g [illegible] gR_2f, [illegible] $f \neq g$.
- [illegible]: [illegible].
  - [illegible]: $Y = \mathbb{N}$, $X = \{1, 2\}$.
  - $f(1) = f(2) = 2$, $g(1) = 3, g(2) = 1$, $h(1) = 1, h(2) = 1$
  - fR_2g ([illegible] 1: $f(1) \leq g(1)$).
  - gR_2h ([illegible] 2: $g(2) \leq h(2)$).
  - [illegible] fR_2h: $f(1) > h(1)$ [illegible] $f(2) > h(2)$.



[illegible]

[illegible]: [illegible] $A = \{0,1,2\}^{\mathbb{N}}$ [illegible]: fRg [illegible]

$$(f = g) \vee (\forall n \in \mathbb{N}(f(n) = f(n)g(n)))$$

[illegible]:

[illegible]. [illegible] R-[illegible] A.

[illegible]. [illegible] A [illegible] R.

[illegible]. [illegible]: R [illegible] A.

[illegible]: [illegible] [illegible] [illegible] [illegible]

[illegible]. R-[illegible] [illegible] [illegible]:

- [illegible]: [illegible] [illegible] ($f = g$).
- [illegible]-[illegible]: [illegible] fRg -[illegible] gRf. [illegible] $f = g$ [illegible]. [illegible], $\forall n(f(n) = f(n)g(n))$ [illegible] $g(n) = g(n)f(n)$. [illegible] [illegible] [illegible], $f(n) = f(n)g(n) = g(n)f(n) = g(n)$, [illegible] $f = g$.
- [illegible]: [illegible] fRg -[illegible] gRh. [illegible] [illegible] [illegible], [illegible]. [illegible], $\forall n(f(n) = f(n)g(n))$ [illegible] $g(n) = g(n)h(n)$. [illegible] [illegible] [illegible]: $f(n) = f(n)g(n) = f(n)(g(n)h(n)) = (f(n)g(n))h(n) = f(n)h(n)$, [illegible] fRh.

[illegible]. [illegible] [illegible]:

- [illegible]: [illegible] [illegible] $\vec{0}$ ([illegible] 0 [illegible] [illegible]). [illegible] [illegible] [illegible].
- [illegible]: [illegible] [illegible] $\vec{1}$ ([illegible] 1 [illegible] [illegible]). [illegible] [illegible] [illegible].

$x \setminus y$	0	1	2
0			
1			
2			

[illegible]. [illegible]:

- [illegible]: R [illegible] [illegible].
- [illegible] $f = (2, 0, 0, \dots)$ -[illegible] $g = (0, 2, 0, \dots)$.
- $f(0) = 2 > 0 = g(0)$ ([illegible] [illegible] [illegible]), [illegible] [illegible] [illegible] fRg.
- $g(1) = 2 > 0 = f(1)$ ([illegible] [illegible] [illegible]), [illegible] [illegible] [illegible] gRf.

[illegible] :[illegible]

[illegible]: $A = \mathcal{P}(\mathbb{N})^{\mathbb{N}}$ [illegible] $\mathbb{N}$-[illegible] $\mathcal{P}(\mathbb{N})$-[illegible]. [illegible] R [illegible] A [illegible]: $\forall n \in \mathbb{N}(f(n) \subseteq g(n)) \iff fRg$.

- [illegible] R [illegible].
- [illegible] R [illegible] ([illegible])? [illegible].
- [illegible] A-[illegible]? [illegible], [illegible]. [illegible], [illegible].
- [illegible] $X = \{f \in A \mid \forall n \in \mathbb{N}(n \in f(n))\}$.

[illegible] X-[illegible]? ([illegible] R). [illegible], [illegible]. [illegible], [illegible].

- [illegible] $D = \{f \in A \mid \forall n \in \mathbb{N}(f(n) \subseteq f(n+1) \wedge f(n) \neq f(n+1))\}$. [illegible] D-[illegible]? [illegible], [illegible]. [illegible], [illegible].

[illegible]: [illegible] [illegible] [illegible] (1 [illegible])

[illegible] R-[illegible] [illegible] [illegible]:

- [illegible]: [illegible] n, $f(n) \subseteq f(n)$, [illegible] fRf.
- [illegible]: [illegible] fRg [illegible]-gRf.
  - [illegible] [illegible] n, $f(n) \subseteq g(n)$ [illegible] $g(n) \subseteq f(n)$.
  - [illegible] [illegible] n, $f(n) = g(n)$.
  - [illegible] $f = g$.
- [illegible]: [illegible] fRg [illegible]-gRh.
  - [illegible] [illegible] n, $f(n) \subseteq g(n)$ [illegible] $g(n) \subseteq h(n)$.
  - [illegible] [illegible] [illegible] $f(n) \subseteq h(n)$.
  - [illegible] fRh.

[illegible] [illegible] [illegible]:

- [illegible] [illegible] [illegible]:
- [illegible] $f(n) = \{1\}$ [illegible]-$g(n) = \{2\}$ [illegible] n.
- $f(0) = \{1\} \not\subseteq \{2\} = g(0)$, [illegible] [illegible] fRg.
- $g(0) = \{2\} \not\subseteq \{1\} = f(0)$, [illegible] [illegible] gRf.

[illegible]: [illegible] (2 [illegible])

[illegible]. [illegible]-A:

- [illegible]. [illegible] $M(n)=\mathbb{N}$ [illegible] n.
- [illegible]: [illegible] $f\in A$ [illegible].
- [illegible] n, $f(n)\in\mathcal{P}(\mathbb{N})$ [illegible] $f(n)\subseteq\mathbb{N}$.
- [illegible] $f(n)\subseteq M(n)$ [illegible] n.
- [illegible] fRM [illegible] f, [illegible]-M [illegible].

[illegible]. [illegible]-X:

- [illegible] X [illegible]-n [illegible]-$f(n)$.
- [illegible]. [illegible] $m(n)=\{n\}$.
- [illegible]:

1. [illegible]-$m\in X$: [illegible] n, [illegible] $n\in m(n)$? [illegible], [illegible] $m(n)=\{n\}$.
2. [illegible]: [illegible] $f\in X$ [illegible].
3. [illegible] X, [illegible] n [illegible] $n\in f(n)$.
4. [illegible] $\{n\}\subseteq f(n)$, [illegible] $m(n)\subseteq f(n)$.
5. [illegible] mRf [illegible] $f\in X$.

[illegible]: [illegible] (3 [illegible])

[illegible]. [illegible] D-[illegible]:

- [illegible] D [illegible].
- [illegible].
- [illegible]: [illegible] $m \in D$.
- [illegible] $f \in D$, [illegible] mRf, [illegible] n, $m(n) \subseteq f(n)$.
- [illegible], $m(n) \subseteq \bigcap\{f(n) \mid f \in D\}$.
- [illegible] D-[illegible]: [illegible] $k \in \mathbb{N}$, [illegible] $f_k(n) = \{k, k+1, \ldots, k+n\}$.
  - [illegible] $f_k \in D$-[illegible]: $|f_k(n)| = n+1$ [illegible]-$|f_k(n+1)| = n+2$, [illegible]- $f_k(n) \subseteq f_k(n+1)$, [illegible].
- [illegible] n [illegible]: [illegible] $x \in \mathbb{N}$, [illegible] $k > x$. [illegible] $x \notin \{k, \ldots, k+n\} = f_k(n)$.
- [illegible] x [illegible] $f_k(n)$-[illegible]. [illegible].
- [illegible] $m(n) \subseteq \emptyset$, [illegible] $m(n) = \emptyset$ [illegible] n.
- [illegible] $m(n) = \emptyset$ [illegible] n, [illegible] $m(n) = m(n+1)$, [illegible] $m \notin D$ ([illegible]).
- [illegible]: [illegible] D-[illegible].

[illegible]: [illegible] [illegible] [illegible] [illegible] ([illegible] 24)

.6 ([illegible] 24) [illegible] R [illegible] f [illegible] $\mathbb{Z}$-[illegible] $\mathrm{dom}(f) = \mathbb{Z}$:

[illegible] $m \in \mathbb{Z}$ [illegible] $f(m+1) R f(m)$,

[illegible] $m \in \mathbb{Z}$ [illegible] $n \in \mathbb{N}$ [illegible] $f(n+m+1) R f(m)$.

[illegible]: [illegible] [illegible] [illegible]

[illegible]: [illegible] $m \in \mathbb{Z}$ [illegible] $n \in \mathbb{N}$ [illegible] n ([illegible] m [illegible]).

[illegible] $(n = 0)$: [illegible]"[illegible]: [illegible] $m \in \mathbb{Z}$, $f(0 + m + 1)Rf(m)$, [illegible] $f(m+1)Rf(m)$. [illegible] ([illegible]).

[illegible]: [illegible] n, [illegible] $k \in \mathbb{Z}$ [illegible] $f(n+k+1)Rf(k)$. [illegible] $n+1$: [illegible]"[illegible] $m \in \mathbb{Z}$, $f((n+1)+m+1)Rf(m)$. [illegible]: $f(n+m+2)Rf(m)$.

[illegible] $m \in \mathbb{Z}$. [illegible] $k = m+1$. [illegible] $k \in \mathbb{Z}$. [illegible] ([illegible] m) [illegible] m, [illegible]:

$$f(m+1)Rf(m)$$

[illegible] k ([illegible]), [illegible] $f(n+k+1)Rf(k)$. [illegible] $k = m+1$:

$$f(n+(m+1)+1)Rf(m+1) \implies f(n+m+2)Rf(m+1)$$

[illegible]:

1. $f(n+m+2)Rf(m+1)$ ([illegible]).
2. $f(m+1)Rf(m)$ ([illegible]).

[illegible] R, [illegible]: $f(n+m+2)Rf(m)$

[illegible], [illegible] $n \in \mathbb{N}$ [illegible] $m \in \mathbb{Z}$.

([illegible] 25) [illegible] [illegible]-[illegible] [illegible] :[illegible]

.7 ([illegible] 25) [illegible] [illegible] [illegible] [illegible] B [illegible] [illegible] [illegible] $\mathcal{A}$ ([illegible] [illegible] 2 [illegible]), [illegible]:

$$B \setminus \left(\bigcap \mathcal{A}\right) = \bigcup\{(B \setminus A) \mid A \in \mathcal{A}\}$$

[illegible]: [illegible]-[illegible] [illegible]

[illegible] $n = |\mathcal{A}|$:

[illegible] ($n = 2$): [illegible] $\mathcal{A} = \{A_1, A_2\}$. [illegible] $\bigcap \mathcal{A} = A_1 \cap A_2$. [illegible] $(B \setminus A_1) \cup (B \setminus A_2)$. [illegible] $B \setminus (A_1 \cap A_2) = (B \setminus A_1) \cup (B \setminus A_2)$. [illegible]-[illegible].

[illegible]: [illegible] n. [illegible] $\mathcal{A}$ [illegible] $n+1$. [illegible] $A_0 \in \mathcal{A}$. [illegible] $\mathcal{A}' = \mathcal{A} \setminus \{A_0\}$. [illegible] $|\mathcal{A}'| = n$, [illegible] $\bigcap \mathcal{A} = (\bigcap \mathcal{A}') \cap A_0$. [illegible] (2 [illegible] $X = \bigcap \mathcal{A}'$ [illegible]-$Y = A_0$):

$$B \setminus (X \cap Y) = (B \setminus X) \cup (B \setminus Y)$$

[illegible]:

$$B \setminus \left(\bigcap \mathcal{A}\right) = \left(B \setminus \bigcap \mathcal{A}'\right) \cup (B \setminus A_0)$$

[illegible] $\mathcal{A}'$:

$$B \setminus \bigcap \mathcal{A}' = \bigcup \{(B \setminus A) \mid A \in \mathcal{A}'\}$$

[illegible]:

$$\left(\bigcup \{(B \setminus A) \mid A \in \mathcal{A}'\}\right) \cup (B \setminus A_0) = \bigcup \{(B \setminus A) \mid A \in \mathcal{A}\}$$

[illegible]: [illegible] (25 [illegible])

8. (25 [illegible]) [illegible] A, [illegible]:

$$|\mathcal{P}(A)| = 2^{|A|}$$

[illegible]: [illegible]

[illegible] $|A| = n$:

[illegible] ($n = 0$): [illegible] 0 [illegible] $A = \emptyset$. [illegible] $\mathcal{P}(\emptyset) = \{\emptyset\}$. [illegible] $\mathcal{P}(\emptyset)$-[illegible] 1. [illegible] $2^0 = 1$. [illegible].

[illegible]: [illegible] n, [illegible] A [illegible] $n+1$. [illegible] A [illegible] $n+1$ [illegible]. [illegible] $x \in A$. [illegible] $A' = A \setminus \{x\}$. [illegible] A' [illegible] n. [illegible] A ([illegible] $\mathcal{P}(A)$-[illegible]) [illegible]:

1. [illegible] [illegible] x: [illegible] A'. [illegible] 2^n [illegible].
2. [illegible] [illegible] x: [illegible] $S \cup \{x\}$ [illegible] $S \subseteq A'$. [illegible] A', [illegible] 2^n [illegible].

[illegible]:

$$|\mathcal{P}(A)| = 2^n + 2^n = 2 \cdot 2^n = 2^{n+1}$$

[illegible]: [illegible] ([illegible])

8. ([illegible] 25) [illegible]: [illegible] $n \geq 2$ [illegible] ([illegible]).

[illegible]: [illegible]

[illegible]: [illegible] n. [illegible]: [illegible] $n = 2$, [illegible] 2 [illegible], [illegible] ([illegible]). [illegible]: [illegible] [illegible] k [illegible] $2 \le k < n$. [illegible] n. [illegible]:

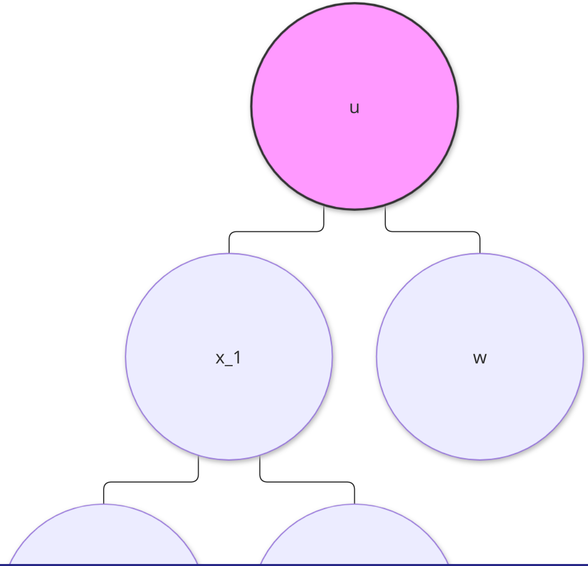
1. [illegible] n [illegible]: [illegible] ([illegible]).
2. [illegible] n [illegible]: [illegible] $a, b \in \mathbb{N}$ [illegible] $n = a \cdot b$ [illegible] $1 < a, b < n$.

- [illegible] $2 \le a < n$, [illegible], a [illegible].
- [illegible] $2 \le b < n$, [illegible], b [illegible].
- [illegible] $a \cdot b$ ([illegible] n) [illegible]. ■

[illegible]: [illegible] ([illegible])

9. ([illegible] 25) [illegible] $G=(V,E)$ [illegible] $E \subseteq V \times V$ [illegible] $E \cap Id_V = \emptyset$ ([illegible]). [illegible] V-[illegible] [illegible] n.
[illegible] $u,v \in V$ [illegible] [illegible] $x_1,\dots,x_k$ ([illegible]) [illegible]: $x_1=u, x_k=v$ [illegible] $1 \le i < k$ [illegible] $(x_i,x_{i+1}) \in E$. [illegible] $|E| = 2(n-1)$.

([illegible]: [illegible]).



[illegible]: [illegible]

[illegible] n [illegible] (V [illegible]).

[illegible]: [illegible] $n = 1$, [illegible] $V = \{v\}$. [illegible] E [illegible] $E \cap I_V = \emptyset$, [illegible] (v, v), [illegible] $E = \emptyset$. [illegible] 0, [illegible] $2(1-1) = 0$.

[illegible]: [illegible] $k < n$. [illegible] n ($n > 1$). [illegible] $(u, v) \in E$ [illegible]. [illegible] E-[illegible] (u, v) [illegible] (v, u).

- [illegible] V [illegible] V_1, V_2 [illegible]:
  - [illegible] V_1 [illegible] u-[illegible], [illegible] V_2 [illegible] v-[illegible].
  - [illegible] ([illegible] u-[illegible], [illegible] v).
  - [illegible] ([illegible] u [illegible] v-[illegible]).
  - [illegible] ([illegible] u [illegible] v-[illegible]).
  - [illegible] "[illegible]" [illegible].
- [illegible] $|V_1| = k$ [illegible]-$|V_2| = m$. [illegible] $k + m = n$, [illegible] $k, m \geq 1$.
- [illegible], [illegible] E-[illegible] V_1-[illegible] $2(k-1)$, [illegible]-V_2 [illegible] $2(m-1)$.
- [illegible] E [illegible]:

$$|E| = 2(k-1) + 2(m-1) + 2 = 2k - 2 + 2m - 2 + 2 = 2(k+m) - 2 = 2n - 2 = 2(n-1)$$

[illegible] :[illegible] [illegible] ([illegible] [illegible])

[illegible]: [illegible] $n \geq 12$ [illegible]-$n = 4a + 5b$ [illegible] $a, b \in \mathbb{N}$.

[illegible]: [illegible] [illegible] ([illegible] [illegible])

[illegible]: [illegible] n.

[illegible]: [illegible] $12 \leq n \in \mathbb{N}, n$. [illegible] k [illegible] $12 \leq k < n$, [illegible] k [illegible] $k = 4a + 5b$ [illegible] $a, b \in \mathbb{N}$. [illegible] n.

[illegible] ([illegible]):

- [illegible] $n = 12$: $12 = 4 \cdot 3 + 5 \cdot 0$ ([illegible] $a = 3, b = 0$).
- [illegible] $n = 13$: $13 = 4 \cdot 2 + 5 \cdot 1$ ([illegible] $a = 2, b = 1$).
- [illegible] $n = 14$: $14 = 4 \cdot 1 + 5 \cdot 2$ ([illegible] $a = 1, b = 2$).
- [illegible] $n = 15$: $15 = 4 \cdot 0 + 5 \cdot 3$ ([illegible] $a = 0, b = 3$).

[illegible] :[illegible]

[illegible] :($n \geq 16$ [illegible]) [illegible] $n \geq 16$, [illegible] $n-4$. [illegible] $12 \leq n-4 < n$ ([illegible] $n \geq 16$, [illegible] $n-4 \geq 12$, [illegible] [illegible]-n).

[illegible], [illegible] $n-4$:

$$n - 4 = 4a' + 5b'$$

[illegible] $a', b' \in \mathbb{N}$.

[illegible] 4 [illegible]:

$$n = (n-4) + 4 = (4a' + 5b') + 4 = 4(a'+1) + 5b'$$

[illegible] $a = a' + 1$ [illegible]-$b = b'$. [illegible]-$a' \in \mathbb{N}$, [illegible] $a \in \mathbb{N}$. [illegible] $b \in \mathbb{N}$. [illegible] $n = 4a + 5b$. ■

[illegible]: [illegible]

- [illegible] $\aleph_0 , 2^{\aleph_0}$ [illegible].
  - $A = \mathbb{Q}^{\mathbb{N}}$
  - $B \subseteq \{0,1\}^{\mathbb{N}}$ [illegible] $\{0,1\}$.
  - $C \subseteq \mathcal{P}(\mathbb{N})^{\mathbb{Q}}$ [illegible].
- [illegible] $D \subseteq \mathbb{N}^{\mathbb{N}}$ [illegible].
  - [illegible] D [illegible] $h : \{0,1\}^{\mathbb{N}} \to D$ [illegible].
  - [illegible] $|D| = \aleph$.

(1 [illegible]) [illegible] :[illegible]

ב. [illegible]:

i. [illegible] $A = \mathbb{Q}^{\mathbb{N}}$:

- [illegible] $|A| = |\mathbb{Q}|^{|\mathbb{N}|} = \aleph_0^{\aleph_0}$.
- [illegible]-ש $2 \le \aleph_0 \le 2^{\aleph_0}$, [illegible]:

$$2^{\aleph_0} \le \aleph_0^{\aleph_0} \le (2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0 \cdot \aleph_0} = 2^{\aleph_0}$$

- [illegible]-[illegible], $|A| = 2^{\aleph_0} = \aleph$.

ii. [illegible] B ([illegible]):

- $B \subseteq \{0,1\}^{\mathbb{N}}$, [illegible] $|B| \le 2^{\aleph_0}$.
- [illegible], [illegible] 2 [illegible] : [illegible] 0 [illegible] 1.
- [illegible] $|B| = 2^{\aleph_0} - 2 = 2^{\aleph_0} = \aleph$.
- ([illegible]: [illegible]).

iii. [illegible] C ([illegible]):

- [illegible] ה"[illegible].
- [illegible].
- [illegible]: $|\mathbb{Q}| = \aleph_0$.
- [illegible]: $|\mathcal{P}(\mathbb{N})| = 2^{\aleph_0} = \aleph$.
- [illegible] ($\aleph_0 \neq \aleph$), [illegible].
- [illegible] C [illegible].
- $|C| = 0$ ([illegible]).

[illegible]: [illegible] (2 [illegible])

[illegible] D [illegible]:

i. [illegible] $h:\{0,1\}^{\mathbb{N}} \to D$:

- [illegible] $f \in \{0,1\}^{\mathbb{N}}$, [illegible] $g_f \in D$ [illegible] $\mathbb{N}$-[illegible] $\mathbb{N}$-[illegible] ([illegible]).
- [illegible]: [illegible] n [illegible] n [illegible] $f(n)$. [illegible] $g_f(n) = 2n + f(n)$.
- [illegible] $f(n) \in \{0,1\}$, [illegible] $2n + f(n)$ [illegible] $2n$ [illegible] $2n+1$.
- [illegible] g_f [illegible]: [illegible] $g_f(n) = g_f(m)$.
  - [illegible] $2n + f(n) = 2m + f(m)$.
  - [illegible] $f(n), f(m) \in \{0,1\}$, [illegible] $f(n)$ [illegible] $f(m)$. [illegible] $f(n) = f(m)$.
  - [illegible] $n = m$ [illegible] $2n = 2m \implies n = m$. [illegible] $g_f \in D$.

- [illegible] h ($f \mapsto g_f$) [illegible]:
  - [illegible] $h(f_1) = h(f_2)$, [illegible] g_{f_1} [illegible] g_{f_2}.
  - [illegible] n, $g_{f_1}(n) = g_{f_2}(n)$.
  - [illegible] $2n + f_1(n) = 2n + f_2(n)$.
  - [illegible] $2n$ [illegible] $f_1(n) = f_2(n)$ [illegible] n.
  - [illegible] $f_1 = f_2$.

ii. [illegible] $|D| = \aleph$:

- [illegible] $D \subseteq \mathbb{N}^{\mathbb{N}}$, [illegible] $|D| \le |\mathbb{N}^{\mathbb{N}}| = \aleph_0^{\aleph_0} = 2^{\aleph_0} = \aleph$.
- [illegible] $\{0,1\}^{\mathbb{N}}$-[illegible] D-[illegible].
- [illegible] $|D| \ge |\{0,1\}^{\mathbb{N}}| = 2^{\aleph_0} = \aleph$.
- [illegible], $|D| = \aleph$.

[illegible]: [illegible] [illegible] [illegible]

[illegible]: A [illegible]. [illegible] A-[illegible] [illegible] [illegible] [illegible] Q [illegible] A [illegible] 2-[illegible] $|Q| = 2$ [illegible] $X \in Q$, $X \sim A$.

[illegible]:

- ([illegible]) [illegible] $\mathbb{N}$-[illegible] [illegible] [illegible].
- ([illegible]) [illegible]: [illegible] A [illegible], [illegible] $\mathcal{P}(A)$ [illegible] [illegible].
- ([illegible]) [illegible] A-[illegible] [illegible]. [illegible]:
  1. A [illegible].
  2. [illegible] B, [illegible] $B \sim A$ [illegible] B [illegible] [illegible].
  3. [illegible] $n \in \mathbb{N} \setminus \{0\}$, [illegible] [illegible] Q_n [illegible] A [illegible] n-[illegible] $|Q_n| = n$ [illegible] [illegible] $X \in Q_n$, $X \sim A$.
  4. $\{B \mid B \in \mathcal{P}(A) \wedge B \sim A\} \sim \mathcal{P}(A)$.

[illegible]

[illegible] $\mathbb{N}$-[illegible]:

- [illegible] $\mathbb{N}$ [illegible] $\mathbb{N}$-[illegible].
- [illegible] $E = \{2k \mid k \in \mathbb{N}\}$ [illegible] $O = \{2k+1 \mid k \in \mathbb{N}\}$.
- [illegible] $\mathbb{N}$, [illegible] $Q = \{E, O\}$ [illegible] $|Q| = 2$.
- [illegible] $f(n) = 2n$ [illegible] $E \sim \mathbb{N}$. [illegible] $g(n) = 2n + 1$ [illegible] $O \sim \mathbb{N}$.
- [illegible] $\mathbb{N}$ [illegible].

□□□□□□: □□□□□□□ □□□□□□□ □□□□□□□ - □□□□□ □'

□. □□ A □□□□□□□□□□ □□ $\mathcal{P}(A)$ □□□□□□ □□□□□□□:

- □□□□□ $X = \mathcal{P}(A)$. □□□□□□ A-□ □□□□□□□□□, □□ □□ □□□□□ $a \in A$.
- □□□□□□ □□□□ □□□□□□□ □□ □□□□-□□□□□□:
  - $S_1 = \{B \in \mathcal{P}(A) \mid a \in B\}$ (□□ □□□□□□□□ □□□□□□□□ □□ a).
  - $S_0 = \{B \in \mathcal{P}(A) \mid a \notin B\}$ (□□ □□□□□□□□ □□□□ □□□□□□□ □□ a).
- □□□□□ $S_1 \cap S_0 = \emptyset$-□ $S_1 \cup S_0 = \mathcal{P}(A)$. □□□□ □□ □□□□□□.
- □□□□□□ S_0: □□□□□ □□□□□□ □□□□□□ □□□□□□ □□ $A \setminus \{a\}$. □□□□□□□ A-□ □□□□□□□□□, $A \sim A \setminus \{a\}$, □□□□□ $S_0 = \mathcal{P}(A \setminus \{a\}) \sim \mathcal{P}(A)$.
  - □□□□□□ $A \sim A \setminus \{a\}$-□:
    - □□□□□□ A-□ □□□□□□□□□, □□ □□ □□□□□-□□ □□ □□□□□□ $A \supseteq B = \{b_0, b_1, b_2, \dots\}$, □□□□□□ □□□□□□ $a = b_0$-□.
    - □□□□□□ □□□□□□□□ $h : A \to A \setminus \{a\}$ □□:
    - □□ $x \notin B$, □□□ $h(x) = x$.
    - □□ $x = b_n \in B$, □□□ $h(b_n) = b_{n+1}$.
    - □□□□□□□□□ h □□□□ □"□□ □□□□ (□□□□□ □□ □□□□□□□□ □-B, □□□□□ □□□□□□ □□□□□). □□□ $|A| = |A \setminus \{a\}|$.
- □□□□□□ S_1: □□□□□□□□□□ $f : S_0 \to S_1$ □"□ $f(B) = B \cup \{a\}$ □□□□□ □□□□□□ (□□"□ □□□□). □□□ $S_1 \sim S_0 \sim \mathcal{P}(A)$.

[illegible]: [illegible] - 1.[illegible]

[illegible]-A [illegible] :1.[illegible]

- [illegible] -A [illegible], [illegible] X, Y [illegible] -$X \cup Y = A$, $X \cap Y = \emptyset$, [illegible]-$X \sim A, Y \sim A$.
- [illegible] -A [illegible]. [illegible] $|A| = n$.
- [illegible] $|X| = n$ [illegible]-$|Y| = n$.
- [illegible], $|A| = |X| + |Y| = n + n = 2n$.
- [illegible] $n = 2n$, [illegible] $n = 0$.
- [illegible] $A = \emptyset$, [illegible] 2-[illegible] ([illegible] $|Q| = 2$ [illegible], [illegible]
  [illegible]).
- [illegible]. [illegible] A [illegible].

2.[illegible] [illegible] - [illegible] [illegible] [illegible] :[illegible]

[illegible] [illegible] [illegible] .2.[illegible]:

- [illegible] B [illegible] $A \sim B$. [illegible] $f: B \to A$ [illegible].
- [illegible] $\{A_1, A_2\}$ [illegible] A [illegible] ($A_i \sim A$).
- [illegible] $B_1 = f^{-1}[A_1]$ [illegible]-$B_2 = f^{-1}[A_2]$.
- [illegible], [illegible] B_1, B_2 [illegible] B.
- [illegible] f [illegible] B_1-[illegible] A_1-[illegible], [illegible] $B_1 \sim A_1$.
- [illegible]: $B \sim A \sim A_1 \sim B_1$. [illegible] $B \sim B_2$.
- [illegible] B [illegible].

3.[illegible] [illegible] - [illegible] :[illegible]

3.[illegible] [illegible] n-[illegible]:

- [illegible] n.
- [illegible] ($n = 1$): [illegible] $\{A\}$ [illegible]. [illegible] ($n = 2$): [illegible] A-[illegible].
- [illegible]: [illegible] n-[illegible]. [illegible] $\{X_1, ..., X_n\}$ [illegible] $X_i \sim A$.
- [illegible] X_1. [illegible] $X_1 \sim A$-[illegible] A-[illegible], [illegible] 2.[illegible] X_1 [illegible].
- [illegible] $\{Y, Z\}$ [illegible] X_1 [illegible] $Y \sim X_1 \sim A$-[illegible] $Z \sim X_1 \sim A$.
- [illegible]: $Q_{n+1} = \{Y, Z, X_2, ..., X_n\}$.
- [illegible] $n + 1$ [illegible], [illegible] ([illegible] $Y, Z \subset X_1$ [illegible]), [illegible] A-[illegible].

4.[illegible] [illegible] - [illegible] [illegible] [illegible] :[illegible]

4.[illegible] [illegible] [illegible]-[illegible] [illegible] A-[illegible]:

- [illegible] $\mathcal{S} = \{B \in \mathcal{P}(A) \mid B \sim A\}$.
- [illegible] $\mathcal{S} \subseteq \mathcal{P}(A)$-[illegible], [illegible] $|\mathcal{S}| \leq |\mathcal{P}(A)|$.
- [illegible]:
- [illegible] $A = X \cup Y$ [illegible] $X \sim A, Y \sim A$.
- [illegible] $Z \subseteq Y$, [illegible] $\psi(Z) = X \cup Z$.
- [illegible] $X \cup Z$ [illegible] X [illegible] $|X \cup Z| \geq |X| = |A|$. [illegible] $|A|$. [illegible] $X \cup Z \in \mathcal{S}$.
- [illegible] ψ [illegible]: [illegible] $X \cup Z_1 = X \cup Z_2$.
  - [illegible] Y: $(X \cup Z_1) \cap Y = (X \cup Z_2) \cap Y$
  - [illegible]: $(X \cap Y) \cup (Z_1 \cap Y) = (X \cap Y) \cup (Z_2 \cap Y)$
  - [illegible] $X \cap Y = \emptyset$-[illegible] $Z_1, Z_2 \subseteq Y$-[illegible] ([illegible] $Z_i \cap Y = Z_i$), [illegible]: $\emptyset \cup Z_1 = \emptyset \cup Z_2 \implies Z_1 = Z_2$
- [illegible] $|\mathcal{S}| \geq |\mathcal{P}(Y)| = |\mathcal{P}(A)|$ ([illegible] $Y \sim A$).
- [illegible].

:

- $A(f,g)=\{n\mid n\in\mathbb{N}\wedge f(n)>g(n)\}$.
- $\mathbb{N}^{\mathbb{N}}$: R $R=\{\langle f,g\rangle\mid f,g\in\mathbb{N}^{\mathbb{N}}\wedge A(f,g)\in Fin(\mathbb{N})\}$.
- $D(f,g)=\{n\mid n\in\mathbb{N}\wedge f(n)\neq g(n)\}$.
- $R^{*}=\{\langle f,g\rangle\mid f,g\in\mathbb{N}^{\mathbb{N}}\wedge D(f,g)\in Fin(\mathbb{N})\}$.

:

- () R - $\mathbb{N}^{\mathbb{N}}$ R " $\mathbb{N}^{\mathbb{N}}$ R .
- () $f\in\mathbb{N}^{\mathbb{N}}$, $R[\{f\}]\sim\mathcal{P}(\mathbb{N})$.
- () $\bigcup\{\mathbb{N}^{A}\mid A\in Fin(\mathbb{N})\}$.
- () $f\in\mathbb{N}^{\mathbb{N}}$, $[f]_{R^{*}}\sim\mathbb{N}$.
- () $\mathbb{N}^{\mathbb{N}}/R^{*}$.
- () $f,g\in\mathbb{N}^{\mathbb{N}}$, $[g]_{R^{*}}$ $[f]_{R^{*}}$ $\underline{R}_{R^{*}}$.

[illegible]: '[illegible] – [illegible] R

1. R [illegible] ([illegible]):

- [illegible]: f [illegible], $A(f,f)=\{n \mid f(n)>f(n)\}=\emptyset$. [illegible], fRf.
- [illegible]: fRg [illegible]-gRh.
  - [illegible] $K_1=\{n \mid f(n)>g(n)\}$ -[illegible] $K_2=\{n \mid g(n)>h(n)\}$ [illegible].
  - [illegible]: $(f(n)>g(n)) \vee (g(n)>h(n)) \implies f(n)>h(n)$.
    - [illegible]: [illegible] $f(n)\leq g(n)$ [illegible] $g(n)\leq h(n)$.
    - [illegible] $f(n)\leq h(n)$, [illegible] -$f(n)>h(n)$.
  - [illegible] $A(f,h)\subseteq K_1\cup K_2$.
  - [illegible], [illegible] $A(f,h)$ [illegible], [illegible] fRh.

2. R [illegible] ([illegible]):

- [illegible] $f(n)=1$ -[illegible] $g(n)=2$ [illegible] n.
- $f(n)$ [illegible] -$g(n)$ [illegible] $A(f,g)=\emptyset$ ([illegible]) $\Leftarrow fRg$.
- $g(n)$ [illegible] -$f(n)$ [illegible] $A(g,f)=\mathbb{N}$ ([illegible]) $\Leftarrow$ [illegible] gRf.
- [illegible]

3. R [illegible]-[illegible] ([illegible])

- [illegible] $f(n)=1, g(n)=1$ [illegible] $n>0$, -[illegible] $f(0)=1, g(0)=2$.
- $fRg \implies A(f,g)=\emptyset$.
- $gRf \implies A(g,f)=\{0\}\in Fin(\mathbb{N})$.
- [illegible] $f\neq g$.

$R[\{f\}]$ [illegible] - '[illegible] [illegible] :[illegible]

[illegible]: [illegible] $f \in \mathbb{N}^{\mathbb{N}}$, [illegible] $R[\{f\}] = \{g \mid fRg\}$ [illegible] [illegible] $\mathcal{P}(\mathbb{N})$-[illegible] ([illegible] $\aleph$).

[illegible]:

- [illegible] $\aleph = 2^{\aleph_0} = |\mathcal{P}(\mathbb{N})|$ [illegible]-$\aleph = |\mathbb{N}^{\mathbb{N}}|$.
- [illegible] [illegible]-$R[\{f\}] \subseteq \mathbb{N}^{\mathbb{N}}$, [illegible] $|R[\{f\}]| \leq \aleph$.
- [illegible] [illegible]-$|R[\{f\}]| \geq \aleph$.
- [illegible] $h \in \mathbb{N}^{\mathbb{N}}$, [illegible] $g_h = f + h$ ([illegible] $g_h(n) = f(n) + h(n)$).
- [illegible] n, [illegible] $g_h(n) \geq f(n)$, [illegible] [illegible] $\{n \mid f(n) > g_h(n)\}$ [illegible] [illegible] ([illegible]).
- [illegible] [illegible] h, [illegible] fRg_h, [illegible] $g_h \in R[\{f\}]$.
- [illegible] $h \mapsto g_h$ [illegible] [illegible]"[illegible] ([illegible] [illegible] $f + h_1 = f + h_2$ [illegible] $h_1 = h_2$).
- [illegible] [illegible] [illegible] $\aleph = |\mathbb{N}^{\mathbb{N}}|$ [illegible] $R[\{f\}]$-[illegible].
- [illegible] [illegible]-[illegible], [illegible] [illegible] $\aleph$.

[illegible] – '[illegible] [illegible]:

[illegible]: [illegible] $S = \bigcup\{\mathbb{N}^A \mid A \in Fin(\mathbb{N})\}$ [illegible] $(\aleph_0)$.

[illegible]:

- [illegible] $Fin(\mathbb{N})$ ([illegible]-[illegible]) [illegible]. ([illegible]).
- [illegible], $\mathbb{N}^A$.
- [illegible] A [illegible], [illegible] $|A| = k$, [illegible] $\mathbb{N}^A$ ([illegible] A-[illegible] $\mathbb{N}$-[illegible]) [illegible] $|\mathbb{N}|^{|A|} = \aleph_0^k = \aleph_0$.
- [illegible], [illegible].
- [illegible]-[illegible]-[illegible].
- [illegible] ("[illegible]"), [illegible] S [illegible] $\aleph_0$.

[illegible] $[f]_{R^*}$

[illegible]: [illegible] $f \in \mathbb{N}^{\mathbb{N}}$, [illegible] $\mathbb{N} \sim [f]_{R^*}$.

[illegible]:

- [illegible], $D(f,g) \in Fin(\mathbb{N}) \iff g \in [f]_{R^*}$.
- [illegible], g [illegible] f-[illegible] [illegible].
- [illegible] S [illegible] $[f]_{R^*}$-[illegible].
- [illegible] $h \in \mathbb{N}^A$ [illegible] $A \in Fin(\mathbb{N})$), [illegible] $\hat{h} \in \mathbb{N}^{\mathbb{N}}$:

$$\hat{h}(n) = \begin{cases} h(n) & n \in A \\ f(n) & n \notin A \end{cases}$$

- [illegible] $\hat{h}$-[illegible] f-[illegible] A [illegible], [illegible] $\hat{h} \in [f]_{R^*}$.
- [illegible] "[illegible]" $[f]_{R^*}$: [illegible] g [illegible] f-[illegible] A, [illegible] h [illegible] g [illegible] A-[illegible].
- [illegible] $|S| = \aleph_0$-[illegible]. [illegible] $|[f]_{R^*}| \leq \aleph_0$.
- [illegible], [illegible] f-[illegible] ($f(0)+k$ [illegible]), [illegible] $|[f]_{R^*}| \geq \aleph_0$.
- [illegible] $\aleph_0$.

[illegible]: [illegible] – [illegible]

[illegible]: [illegible] $\mathbb{N}^{\mathbb{N}}/R^*$ [illegible].

[illegible]:

- [illegible] $X = \mathbb{N}^{\mathbb{N}}$. [illegible] $\aleph = |X|$.
- [illegible] X [illegible] X.

$$X = \bigcup_{C \in X/R^*} C$$

- [illegible] C [illegible] ($|C| = \aleph_0$).
- [illegible]. [illegible]"[illegible].
- [illegible] $\aleph_0$.
- [illegible] $\aleph = |X|$ ([illegible] $\aleph_0$-[illegible]).
- [illegible].
- [illegible]: [illegible] ([illegible] $\aleph$).

[illegible]: [illegible] – [illegible]

[illegible]: [illegible] $[f],[g]$ [illegible] $[f]<[g]$ ([illegible]), $[g]$ [illegible] $[f]$. [illegible]: [illegible] h [illegible] $[f]<[h]<[g]$.

- [illegible] $P=\{n\in\mathbb{N}\mid g(n)>f(n)\}$. [illegible] P [illegible].
- [illegible] P [illegible] P_1,P_2 ([illegible], [illegible] P).
- [illegible] h:

$$h(n)=\begin{cases} g(n) & n\in P_1 \\ f(n) & n\notin P_1 \end{cases}$$

- [illegible] $[f]<[h]$:
  - $f(n)\le h(n)$ [illegible] n ([illegible] P_1, $f<g=h$, [illegible] $f=h$). [illegible] $[f]\le[h]$.
  - [illegible] $\{n\mid h(n)>f(n)\}$ [illegible] P_1 ([illegible]), [illegible] hRf. [illegible] $[f]\neq[h]$.
- [illegible] $[h]<[g]$:
  - $h(n)\le g(n)$ [illegible] n ([illegible] P_1 [illegible], [illegible] $h=f\le g$). [illegible] $[h]\le[g]$.
  - [illegible] $\{n\mid g(n)>h(n)\}$ [illegible] P_2 ([illegible]), [illegible] $[g]\le[h]$.
- [illegible] $[f]<[h]<[g]$. [illegible].

[illegible]: [illegible]

- [illegible] $g : A \to B, f : B \to C$ [illegible].
- [illegible] $f \circ g$ [illegible], [illegible] g [illegible].
- [illegible] $f \circ g$ [illegible], [illegible] f [illegible].
- [illegible] $f : A \to B$ [illegible].
- [illegible] f-[illegible] [illegible] C [illegible] $g, h : C \to A$, [illegible] $f \circ g = f \circ h$ [illegible] $g = h$.
- [illegible] f-[illegible] [illegible] f [illegible].

□□□□□□: □□□□□□ □□□□□□□□□□

□. □□□□□□□ □□□□□□□□:

i. □□□□□: $f \circ g$ □□"□.

- □"□: g □□"□.
- □□□□□□: □□□□□ $x_1, x_2 \in A$ □□ $g(x_1) = g(x_2)$-□.
- □□□□□□ □□ f □□□ □□□□□□□: $f(g(x_1)) = f(g(x_2))$.
- □□□ □□□□□□ □□□□□□□: $(f \circ g)(x_1) = (f \circ g)(x_2)$.
- □□□□□ $f \circ g$-□ □□"□, □□□□□ $x_1 = x_2$-□.
- □□□ g □□"□.

ii. □□□□□: $f \circ g$ □□.

- □"□: f □□.
- □□□□□□: □□□ $c \in C$.
- □□□□□ $f \circ g$-□ □□, □□□□□ $a \in A$ □□ $(f \circ g)(a) = c$-□.
- □□□ □□□□□□ □□□□□□□: $f(g(a)) = c$.
- □□□□□ $b = g(a) \in B$. □□□□□□□ $f(b) = c$.
- □□□ □□□ $c \in C$ □□□□□ □□□□□ □-B, □□□□□□ f □□.

□. □□□□□□ □□□□□□ $\iff$ □□"□:

$\Rightarrow$ □□□□□□ □□□□□□: □□□□□ f □□"□.

- □□□□□ $f \circ g = f \circ h$.
- □□□ $c \in C$ □□□□□□. □□□ $f(g(c)) = f(h(c))$.
- □□□□□□□ □-f □□"□, □□□□□ □-$g(c) = h(c)$.
- □□ □□□□□ □□□ c, □□□ $g = h$.

$\Leftarrow$ □□□□□□ □□□: □□□□□ f □□□□□□□□ □□□□□□.

- □□□□□□ □-f □□"□.
- □□□□□ $x, y \in A$ □□ □-$f(x) = f(y)$.
- □□□□□□ □□□□□□ $C = \{k\}$ (□□□□□□□), □□□□□□□□□□□ $g, h : C \to A$ □"□ $g(k) = x, h(k) = y$
- □□□□□□□: $(f \circ g)(k) = f(x) = f(y) = (f \circ h)(k)$.
- □□□ $f \circ g = f \circ h$.
- □□□ □□□□□□ (□□□□□□ □□□□□□), □□□□□ $g = h$.
- □□□ □□□□□ $g(k) = h(k)$, □□□□□□ $x = y$.
- □□□□□□□ □□□ $f(x) = f(y)$ □□ $x = y$, □□□□□□ f □□"□.